CheckerBoard(void) – [testConstructor]

Input: N/A

State: N/A

Output: N/A

State:
board = new char [ROW_NUM][COL_NUM]
pieceCount = new hashmap<Character,Integer>
viableDirections = new HashMap<Character,ArrayList<DirectionEnum>

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

whatsAtPos(BoardPosition pos) – [testWhatsAtPos_MinRow_MinCol_xPiece]

Input: pos = (0,0)

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output: ‘x’

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

whatsAtPos(BoardPosition pos) – [testWhatsAtPos_MinRow_MaxCol_blackTile]

Input: pos = (0,7)

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output: '*'

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

whatsAtPos(BoardPosition pos) – [testWhatsAtPos_MaxRow_MinCol_blackTile]

Input: pos = (7,0)

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output: '*'

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

whatsAtPos(BoardPosition pos) – [testWhatsAtPos_MaxRow_MaxCol_oPiece]

Input: pos = (7,7)

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output: 'o'

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

whatsAtPos(BoardPosition pos) – [testWhatsAtPos_Row5_Col4_blackTile]

Input: pos = (5,4)

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output: '*'

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

placePiece(BoardPosition pos, char player) – [PlacePieceTest_RoutinePlace_Player_o_Row4_Col0]

Input:
pos: (4, 0)
player: ‘o’

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output:
N/A

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
o	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

placePiece(BoardPosition pos, char player) – [PlacePieceTest_RoutinePlace _Player_x_Row3_Col3]

Input:
pos: (3, 3)
player: ‘x’

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output: N/A

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*	x	*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

placePiece(BoardPosition pos, char player) – [PlacePieceTest_EdgeCase _Player_x_Row0_Col0]

Input:
pos = (0,0)
player= ‘x’

State:

	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output:
N/A

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

placePiece(BoardPosition pos, char player) – [PlacePieceTest_EdgeCase _Player_o_Row7_Col7]

Input:
pos = (7, 7)
player = ‘o’

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	

Output:
N/A

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

placePiece(BoardPosition pos, char player) – [PlacePieceTest_Place_x_On_o_Row6_Col0]

Input:
pos: (6, 0)
player: ‘x’

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output:
N/A 5

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
x	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

getPieceCounts(void) – [testGetPieceCounts_startingBoard]

Input: N/A

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output:
pieceCounts = {‘x’ -> 12, ‘o’ -> 12}

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

getViableDirections(void) – [testGetViableDirections_startingBoard]

Input: N/A

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output:
viableDirections = {‘x’-> [SW, SE] ,
‘o’ -> [NE, NW]}

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

checkPlayerWin(Character player) – [checkPlayerWinTest_Player_x_One_x_Zero_o]

Input:
player = ‘x’

State:
pieceCount = <’x’, 1>
<’o’, 0>

	*		*		*		*
*		*		*		*	
	*		*	x	*		*
*		*		*		*	
	*		*		*		*
*		*		*		*	
	*		*		*		*
*		*		*		*	

Output:
True

State: (Unchanged)

	*		*		*		*
*		*		*		*	
	*		*	x	*		*
*		*		*		*	
	*		*		*		*
*		*		*		*	
	*		*		*		*
*		*		*		*	

checkPlayerWin(Character player) – [checkPlayerWinTest_Player_o_One_x_One _o]

Input:
player = ‘o’

State:
pieceCount = <’x’, 1>
<’o’, 1>

	*		*		*		*
*	x	*		*		*	
	*		*		*		*
*		*		*		*	
	*		*		*		*
*	o	*		*		*	
	*		*		*		*
*		*		*		*	

Output:
False

State: (Unchanged)

	*		*		*		*
*	x	*		*		*	
	*		*		*		*
*		*		*		*	
	*		*		*		*
*	o	*		*		*	
	*		*		*		*
*		*		*		*	

crownPiece(BoardPosition posOfPlayer) – [TestCownPiece_Row7_Column1]

Input: posOfPlayer: (7, 1)

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output: n/a

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	O	*	o	*	o	*	o

crownPiece(BoardPosition posOfPlayer) – [TestCrownPiece_Row2_Column0]

Input: posOfPlayer: (2, 0)

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output: N/A

State:

x	*	X	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

crownPiece(BoardPosition posOfPlayer) – [TestCrownPiece_Row5_Column3]

Input: posOfPlayer: (5, 3)

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output: N/A

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	O	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

movePiece(BoardPosition startingPos, DirectionEnum dir) – [movePieceTest_SEMove_x_Row2_Col0]

Input:
startingPos = (2, 0)
dir = SE
State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*

Output:
(3, 1)
State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
	*	x	*	x	*	x	*
*	x	*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

*	o	*	o	*	o	*	o

movePiece(BoardPosition startingPos, DirectionEnum dir) – [movePieceTest_NEMove_'o'_Row5_Column1]

Input: startingPos: (5, 1)
Dir: NE

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output: (4, 2)

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*	o	*		*		*
*		*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

movePiece(BoardPosition startingPos, DirectionEnum dir) – [movePieceTest_SWMove_'x'_Row2_Column4]

Input: startingPos: (2, 4)
Dir: SW
State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output: (3, 3)

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*		*	x	*
*		*	x	*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

jumpPiece(BoardPosition startingPos, DirectionEnum dir) – [TestJumpPiece_NEMove_'o'_Row4_Column2]

Input: startingPos: (4,2)
Dir: NE
State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*		*	x	*
*		*	x	*		*	
	*	o	*		*		*
*		*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output: (2,4)

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	o	*	x	*
*		*		*		*	
	*		*		*		*
*		*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

jumpPiece(BoardPosition startingPos, DirectionEnum dir) – [TestJumpPiece_SWMove_'x'_Row3_Column3]

Input: startingPos: (3, 3)
Dir: SW
State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*		*	x	*
*		*	x	*		*	
	*	o	*		*		*
*		*	o	*	o	*	o
o	*	o	*	o	*	o	*

Output: (5, 1)

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*		*	x	*
*		*		*		*	
	*		*		*		*
*	x	*	o	*	o	*	o
o	*	o	*	o	*	o	*

*	o	*	o	*	o	*	o

*	o	*	o	*	o	*	o

jumpPiece(BoardPosition startingPos, DirectionEnum dir) –
[TestJumpPiece_SEMove_x_Row2_Column0]

Input: startingPos: (2,0)
Dir: SE
State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*	o	*		*		*	
	*		*		*		*
*	o	*		*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output: (4,2)

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
	*	x	*	x	*	x	*
*		*		*		*	
	*	x	*		*		*
*	o	*		*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

scanSurroundingPositions(BoardPosition startingPos) –
[TestScanSurroundingpositions_'o'SWSE_blankNWNE]

Input: startingPos: (5, 1)

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output: {‘o’ -> [SW, SE], ‘ ‘ -> [NW, NE]}

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*		*		*		*	
	*		*		*		*
*	o	*	o	*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

scanSurroundingPositions(BoardPosition startingPos) – [Test ScanSurroundingpositions_’o’SE_’x’NE]

Input: startingPos: (2, 0)

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*	o	*		*		*	
	*		*		*		*
*	o	*		*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output: {‘x’ -> [NE], ‘o’ -> [SE]}

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*	o	*		*		*	
	*		*		*		*
*	o	*		*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

scanSurroundingPositions(BoardPosition startingPos) – [Test ScanSurroundingpositions_’o’SW_’x’NWNE_blankSE]

Input: startingPos: (3, 1)

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*	o	*		*		*	
o	*		*		*		*
*		*		*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

Output: {'x' -> [NW, NE], 'o' -> [SW], ' ' -> [SE]}

State:

x	*	x	*	x	*	x	*
*	x	*	x	*	x	*	x
x	*	x	*	x	*	x	*
*	o	*		*		*	
o	*		*		*		*
*		*		*	o	*	o
o	*	o	*	o	*	o	*
*	o	*	o	*	o	*	o

getDirection(DirectionEnum dir) – [getDirectionTest_SW]

Input: dir = DirectionEnum.SW State: N/A	Output: BoardPosition = new BoardPosition(1, -1) State: N/A

What tests did each team member write? Just tell me the names of the functions (unless for some reason multiple team members wrote functions for the same method. In that case, tell me which tests specifically by giving me the test names)

[member 1] Hayden Lorenz	CheckerBoard (1 Test) whatsAtPos (5 Tests) GetPieceCounts (1 Test) GetViableDirections (1 Test)

[member 2] Nicholas Gagnon	placePiece (5 Tests) Done getDirection (1 Test) Done checkPlayerWin (2 Tests) Done MovePiece (1 Test)
[member 3] Kara Tholen	MovePiece (2 tests) JumpPiece (3 tests) ScanSurroundingPositions (3 tests) CrownPiece (3 tests)
[member 4]	